GOSIM

GOSIM Shenzhen 2026

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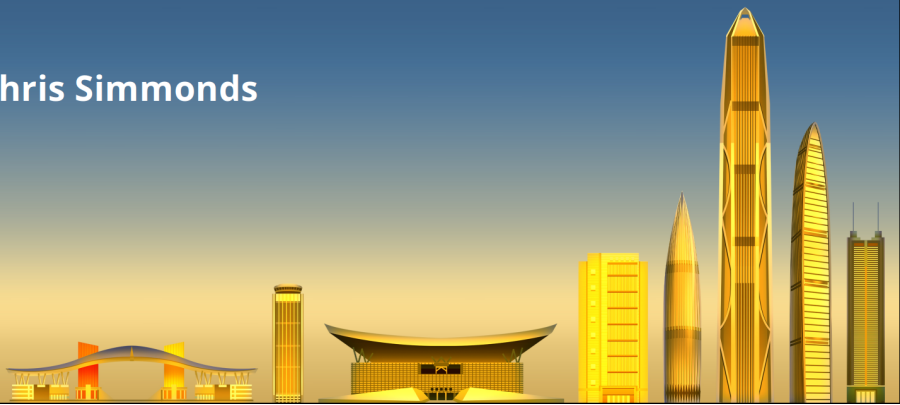


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Which operating system is best for agentic mobile devices?

Chris Simmonds



Overview

- New class of device: "AI companion"
- Form factors are not yet settled, but include
  - AI handheld (e.g. Rabbit R1)
  - memory pendant, e.g. Omi, Plaud NotePin
  - AI glasses, e.g. Ray-Ban Meta, Omi Glass
  - embodied robot, e.g. ElliQ, Loona
- This talk focuses on the handheld category (*)
  - essential hardware
  - options for the operating system






(*) Other form factors are more suited to a small microcontroller paired with a conventional smartphone. I will not consider this class of device

AI handheld device vs smartphone

AI handheld devices present an opportunity to break with the conventional smartphone architecture. Time to rethink

- open source - to promote a string ecosystem among developers
- privacy-controlled
- local memory and offline modes of operation
- replaceable AI providers
- situation-awareness via input from a range of sensors

Do we need apps?

- AI centric => user interaction is primarily through an AI Agent
- Lower dependence on apps, lower dependence on smartphone app ecosystems
- Whatever you want to know or do, the agent can do it for you

Mobile hardware - what's special?

- Battery powered => energy budget is critical
- Highly connected, via:
  - local networks: Wi-Fi, Bluetooth
  - wide area: 4G, 5G
- User interaction via microphone, speaker, touch Screen
- Local AI acceleration to be able perform offline

Common for some of these drivers to be closed source

Especially GPU, NPU and 4G/5G modem

Mobile operating systems and security

Mobile handsets are vulnerable

- Physically, from being lost or stolen
- Remotely, via malicious apps, data or software updates

Respond with "Defence in depth"

- Physical security, rooted in hardware chain of trust
- Execution security, by sandboxing apps and services
- Platform security, through read-only filesystems, signing app packages, signing OTA images

Security has to be validated in CI via test suites such as CTS nd VTS

Open source mobile operating systems

We can split them up into three categories:

- Android
  - Android + Google services
  - AOSP + other service provider
- Linux + Android BSP
  - Sailfish OS
  - Ubuntu Touch
  - Droidian
- Pure Linux
  - Nura (was postmarketOS)
  - Mobian

Obvious choice

Android + Google services

- Google are migrating Android to become an AI-first device
- leaves OEMs little room for innovation

AOSP + third party services

- core development is still controlled by Google
- security updates twice per year unless you become a Google partner
- complex framework supporting legacy apps leaves a large attack surface

Android is only really necessary if you need full Android app support

The need for a Board Support Package

The BSP consists of the drivers and libraries needed to support the SoC and other hardware

A BSP includes:

- Linux kernel
- drivers for NPU, GPU, display, Camera, Audio, GPS, Sensors, Modem and telephony

The BSP is supplied by the SoC vendor and OEM

- most SoC vendors supply only Android BSPs
- consequently, we need tools to run other operating systems on an Android BSP:
  - libhybris
  - Halium

Halium and libhybris trade architectural purity for practical hardware compatibility, while postmarketOS and Mobian generally accept slower device enablement in pursuit of upstream, maintainable Linux support.

https://github.com/libhybris/libhybris

Android has a complex Hardware Abstraction Layer with dependencies on many Android system libraries. All are linked with the Android C library, **bionic**

Typically, libhybris is used to load binary drivers, no source code required

- used to load drivers that link against bionic inside processes whose native C library is glibc or musl
- the driver runs in a stripped down Android system that is launched as a systemd service or container

Halium

https://halium.org/

Halium uses libhybris to create an abstraction layer that runs on top of an Android HAL

The layers are

- Mobile Linux distribution, e.g. Ubuntu Touch, Droidian, etc.
- Interface, apps, system services, including oFono, PulseAudio, Wayland/Mir
- Compatibility bridges: libhybris, libgbinder, camera and audio adaptation components
- Minimal Android environment, usually running in an LXC container
- Android HALs and vendor services
- Android-derived kernel and firmware
- Phone hardware

Sailfish OS

Developed and marketed by Jolla (https://jolla.com/)

Based on libhybris/Halium, on top of an Android BSP

Mixed-source platform

- Open source compons:
  - Qt5 shell, native apps usually written in QML
  - Standard Linux components, including Wayland compositor
- Closed source:
  - Sailfish UI and Jolla applications
  - AppSupport for Android compatibility

Security: Firejail app sandbox with Sailjail app permissions

Sample hardware ports: Jolla and Jolla C2 phone; Xperia 10; plus 12 community ports

Ubuntu Touch

Ubuntu for mobile, developed by UBports https://ubports.com/

Based on libhybris/Halium, on top of an Android BSP

Fully open source

- Qt5 shell, native apps written in QML, JavaScript and C++
- apt installer for Ubuntu packages (warning: most not optimised for phone)
- Android compatibility through Waydroid (see later)

Security: Flatpak (see later)

Sample hardware ports: Vola Phones; Fairphones; OnePlus Nord N100; plus 20 community ports

Droidian (aka hybris-mobian)

Debian for mobile https://droidian.org/

Based on libhybris/Halium, on top of an Android BSP

Features

- GNOME/GTK shell
- deb/apt packaging
- Android compatibility through Waydroid (see later)

Security: Flatpak (see later)

Sample hardware ports: Xiaomi Poco M2 Pro, Sony Xperia 5, Volla Phones

Nura (was PostmarketOS)

Mainline Linux for mobile https://nura.eco/

Features

- Mainline Linux kernel (i.e. does not use libhybris/Halium)
- Based on Alpine Linux
- security based on Duranium https://postmarketos.org/blog/2026/03/17/introducing-duranium/, and on Flatpak isolation
- Duranium: immutable /usr filesystem; with image updates; verified by dm-verity

Security: Flatpak (see later) and Duranium

PostmarketOS an advantage from using mainline kernel: CVEs are fixed more quickly, and there is little or no proprietary code in vendor driver modules

Mobian

Debian for mobile with mainline kernel https://mobian-project.org/

Features

- Mainline Linux kernel (i.e. does not use libhybris/Halium)
- apt installer for Debian packages (warning: most not optimised for phone)
- Phosh (GTK) or Plasma Mobile (QT5) shells

Security: Flatpak (see later)

Sample hardware ports: PinePhone, Oneplus 6/6T

Background: Flatpak

Flatpak is a cross-distribution Linux application format with security features

https://flatpak.org/

- Each application runs inside an isolated sandbox
- Applications include or reference their required runtime dependencies

Security benefits

- Restricts access to user files, devices, processes and system services
- Gives each application its own private data directory
- Uses Linux namespaces and system-call filtering to limit possible damage
- Uses portals to request controlled access - for example, selecting one file instead of exposing the entire home directory
- Allows permissions to be inspected, granted or revoked

Background: Waydroid

Waydroid runs full Android in a container on a GNU/Linux system running Wayland based desktop `https://waydro.id/`

- Uses Linux namespaces rather than hardware virtualisation
- Integrates Android application windows into a Wayland desktop
- Shares the host's Linux kernel and can access hardware acceleration directly

Android compatibility: Android 13, Android API level 33

Background: Phosh and Plasma Mobile

Graphical user environments, both based on Wayland

Phosh (Phone Shell) https://phosh.mobi/

- Uses GTK and adaptive GNOME applications
- Simple, conservative phone interface

Plasma Mobile (https://plasma-mobile.org/

- Uses Qt 5, QML and Kirigami applications
- More configurable and visually feature-rich

Last slide

- Rethinking the mobile operating system
- Each of the options presented has merit
- Most require some work - some require considerable work
- Consequently - there is no winner

Thanks!

